

Danny Muir
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Education

University of Washington, Seattle *Expected June 2027*
Bachelor of Science in Computer Engineering, Paul G. Allen School of Computer Science & Engineering **GPA: 3.84**
Minor: Neural Computation and Engineering **Dean's List:** All Quarters 2023–2025
Honors: Tau Beta Pi Engineering Honor Society
Relevant Coursework: Software Design & Implementation (CSE 331), Machine Learning (CSE 446), Data Structures & Algorithms (CSE 373), Hardware/Software Interface, Digital Circuits, Scientific Computing, Physics, Chemistry, Biology

Experience

Undergraduate Researcher, Neural Systems Laboratory, UW Oct 2025–Present
Advisor: Dr. Rajesh Rao, Paul G. Allen School of Computer Science & Engineering

- Optimizing multi-session neural decoding framework in collaboration with Matthew Bryan, building on temporal basis function models for closed-loop neural stimulation (Bryan et al., arXiv:2507.15274).
- Testing and selecting optimal implementations for session-specific embeddings and test-time adaptation methods.
- Improving training efficiency, model complexity, and cross-session generalization through systematic empirical evaluation.
- Beginning honors thesis research on modeling traveling waves in motor cortex via thalamic signals (Winter 2026).

Associate Researcher, VP Operations, Sensors, Energy, and Automation Laboratory (SEAL), UW Jan 2025–Oct 2025
Advisor: Dr. Alexander Mamishev, Department of Electrical & Computer Engineering

- Conducted VLF/ULF signal processing research for atmospheric and geophysical monitoring applications.
- Co-authored federal grant proposals to DOE, NIH, NSF, CDC, and Google (STTR/SBIR programs).
- Led project management team developing custom automation and DevOps solutions for laboratory operations.

Leadership

Vice President & New Member Educator, Triangle Fraternity, UW Dec 2023–Present

- Manage a STEM-exclusive fraternity, overseeing academic and social initiatives.
- Organize events, study resources, and mentorship programs.
- Update and enhance the educational program for new members, improving integration processes and developmental outcomes.

Activities & Volunteer Coordinator, HOSA, UW Sep 2024–Apr 2025

- Coordinate volunteer opportunities and professional development events for future healthcare professionals.

Skills

Programming: Python, Java, C, JavaScript, Bash, Git, HTML, LaTeX | **Platforms:** Linux, Windows
ML/AI: PyTorch, TensorFlow, NumPy, Pandas, scikit-learn, Neural Networks, Autoencoders, Computer Vision, Large Language Models, Test-Time Adaptation
Neuroscience: EEG/ECOG signal processing, neural decoding models, closed-loop brain-computer interfaces
Research: Grant writing, project management, signal processing, experimental design, technical documentation